Final Practical — Multi-Stakeholder Presentation & Portfolio Guide

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Companion Presentation & Video Files in `docs/`:

- **PowerPoint Presentation (`.pptx`):** `FinDoc_AuditEngine_Presentation_Slides.pptx` (12 Widescreen 16:9 Slides)
- **Landscape Slide Deck (`.pdf`):** `FinDoc_AuditEngine_Slide_Deck.pdf` (12 Designed 16:9 Slides)
- **Recorded Demo Video (`.mp4`):** `FinDoc_AuditEngine_Demo_Walkthrough.mp4` (HD 720p Walkthrough)

Verified Automated Tests: Day 29: `28 passed` (`python -m pytest tests -q`) | Day 30: `33 passed` (`python -m pytest tests -q`) — *all collected tests passed*

1. Tailored Presentation Strategy by Audience (10–12 Minute Structure)

Track A: Presenting to Mentors

- **Focus:** Practical domain motivation, iterative improvements from code review through final polish, and honest engineering reflection.
- **Key Points:**

1. Inspired by real procurement workflows (managing purchase orders and vendor sponsorships for VIT's Riviera & Gravitas festivals).
2. Evolved into a clean, self-contained FastAPI working prototype with an institutional light-theme workbench (`1 USD = 86.50 INR`).
3. Demonstrates three representative scenarios clearly: a clean auto-approved invoice (`INV-1001`), a contract price discrepancy (`INV-1004`), and a high-risk anomaly (`INV-1005`).

Track B: Presenting to Team Members

- **Focus:** Code organization, API contracts, testing workflow, and reproducibility.
- **Key Points:**

1. Modular directory structure separating `app/core` (config, logging, regex privacy scrubber), `app/services` (3-way matcher, anomaly scorer, router), and `app/routers` (`audit`, `mentor_prep`, `portfolio`).
2. Fail-fast `Pydantic v2` schema validators (`@model_validator`) enforcing `$0.05` subtotal tolerances and valid GRN quantities.
3. Reproducible automated test suite (`python -m pytest tests -q` -> `28 passed` in Day 29; `33 passed` in Day 30 — all collected tests passed).

Track C: Presenting to the Technical Panel

- **Focus:** Mathematical logic, synthetic baseline scope, perturbation attribution, and transparent discussion of prototype limitations.
- **Key Points:**

1. **Hybrid Design:** Deterministic 3-way matching enforces configured SKU price/quantity rules, while the 7-feature `IsolationForest` surfaces unusual multi-feature patterns.
2. **Accurate Model Scope:** The `IsolationForest` is trained on a seeded synthetic baseline of 1,000 records (`random_state=42`), and its `0–100` output is an engineered anomaly-risk indicator rather than a calibrated probability of fraud.
3. **Prototype Limitations & Future Work:** Clearly explain that `AUDIT_LEDGER` is an in-memory Python list (requiring PostgreSQL for persistence), `FinancialPrivacyScrubber` and `/parse-raw` use regex rules (requiring NER and optical OCR for production), and model thresholds require real-world calibration.